Dinakaran v

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Portfolio

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Education

- **Nantha College of Technology** — Erode, India
  Bachelor of Engineering in Mechanical; CGPA: 8.0 — *Jul 2018 – Jun 2021*

Skills

Primary Skills: NextJS, ReactJS, JavaScript, TypeScript, HTML 5/CSS, NodeJS, Git

Secondary Skills: ExpressJS, MongoDB, SQL

Experience

- **HCLTech** — Chennai, India
  Lead Engineer — *Agu 2025 – Present*
  - **Training**: Applied newly acquired GenAI knowledge to explore automation opportunities and prototype intelligent assistants for internal tools
- **Bipolar Factory Private Limited** — Chennai, India
  Senior Software Engineer — *Oct 2023 – Jul 2025*
  - **Advanced Next.js Development**: Built a high-performance application using Next.js with SSR/SSG, resulting in faster load times, optimized rendering, and a noticeably smoother user experience.
  - **Advanced Chart System**: Designed and implemented a new charting module using a combination of libraries, delivering highly accurate visuals, dynamic responsiveness, and reusable components capable of handling large datasets.
  - **Custom Widget Framework**: Introduced a drag-and-drop widget system that enabled users to access and organize key dashboard charts quickly, improving usability and dashboard flexibility.
  - **Single Sign-On (SSO) Integration**: Designed and deployed a complete multi-project SSO system that enabled secure role-based access, streamlined authentication with reusable session tokens, and cut repeated login effort by **~20%**.
  - **Redux RTK State Architecture**: Built a centralized global state using Redux Toolkit and RTK Query with smart caching and request deduplication, reducing redundant API calls by **~30%** and significantly improving data flow efficiency.
  - **Runtime Env Injection**: Built a Docker-based runtime environment injection mechanism that allows environment variables to be injected at container startup, eliminating manual config updates and significantly speeding up POC delivery time for Next.js applications.
- **Apphub Technologies LLP** — Chennai, India
  Software Engineer, — *Sep 2022 and Jun 2023*
  - **Custom PDF Compiler**: Developed a custom PDF compiler that renders color-coded charts and dynamic table values directly inside the PDF, improving data clarity and helping users understand visual changes **40% faster** without checking raw tables repeatedly.
  - **Reusable Components & Dynamic Data Flow**: Built a library of reusable React components—such as custom multiselects, dropdowns, lists, and tables—fully driven by dynamic data and flexible styling to support scalable UI development.
  - **Reusable WebSocket Layer**: PEngineered a modular WebSocket abstraction that is reused across the application to deliver real-time data updates and notifications with minimal integration effort.
  - **Frontend Optimization**: Built a custom authentication hook and implemented React optimization patterns, cutting redundant component re-renders by **30%** and boosting UI performance.
- **C Cube Technologies** — Erode, India
  Application engineer — *Jun 2021 – Jun 2022*
  - **User Management Webpage**: Developed a complete user-management interface using React, which strengthened my frontend skills and delivered a more streamlined workflow for internal management operations.
  - **Industrial Automation Projects**: Built multiple automation solutions using ladder logic for **Keyence, Delta, and Siemens PLCs**, and created SCADA visualizations through One-Touch software to support real-time monitoring and control.

Projects

- **OLY Retail/VMS**: An AI retail intelligence system that tracks footfall, customer movement, and in-store trends using existing CCTV cameras.
- **User Management**: Interface within an application that enables admins or managers to manage user accounts, roles, and permissions.
- **World Table Tennis and International Table Tennis Federation**: WTT organizes commercial tournaments while ITTF governs rules and international competition